GlucoTwin — system architecture

A patient-specific type 2 diabetes digital twin. Four input streams are harmonised onto one causal timeline, fused by the model layer, calibrated into prediction intervals and adverse-event probabilities, and served to two clinician interfaces. Raw patient records never leave the clinic.

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1 · SOURCES

Inside the clinic — raw records never leave

STATIC EHR

Health record

Age, sex, BMI, HbA1c, fasting glucose, eGFR, diagnoses, medications, genetic markers. Synthea-ready.

REAL · OPEN

CGM · 145 patients

CGMacros (45 adults, PhysioNet) and Shanghai T2DM (100 patients, figshare). Removing this stream hurts the forecast most of all.

REAL + SIMULATED

Wearable

Fitbit heart rate and activity; sleep, HRV and steps simulated. At +60 min: **+0.04 mg/dL, CI includes zero** — reported, not hidden.

REAL + PRESETS

Meals

Logged macronutrients, plus 16 Indian presets from IFCT-2017. Meals are the **largest single gain**.

2 · HARMONISE & FUSE

One causal timeline

Common grid. Every cohort resampled to a single 5-minute schema with a shared column set; loader checksums verified.

Causal features only. No feature may use information from after the prediction instant — enforced in code and covered by tests.

Patient-grouped splits. Training, inner-validation, calibration and holdout patients never overlap, so no patient appears on both sides of an evaluation.

Gap and quality handling. Sensor gaps flagged rather than silently interpolated; a per-window quality score travels with the forecast.

Fused feature set. Recent CGM history and its slope · time of day · meal macronutrients · heart rate and activity · the static profile.

`src/data · src/features`

3 · MODEL LAYER

Fusion forecaster

Tuned LightGBM on the *change* from the current value at +30, +60 and +120 min. The served model.

Mechanistic twin

A per-patient glucose–insulin ODE. RK4 integration, extended Kalman filter state. Also supplies the features used by the hybrids.

Hybrid physics + ML

Twin-derived features fed to the forecaster, and a GRU trained with an ODE-consistency loss.

Personalisation

Fine-tuning or recursive least squares on the patient's own days: about 1 mg/dL better after 3 to 5 days.

Federated training

FedAvg and FedProx across simulated clinics. Five-clinic and two-site layouts, plus a privacy noise sweep.

4 · CALIBRATE

80% intervals

Split-conformal and CV+. Coverage 80.8% (CGMacros) and 79.5% (Shanghai) at +60 min, per subgroup as well as pooled.

Event probabilities

Platt and isotonic calibration for crossing 180, 140 and 70 mg/dL. Calibrated means 40% happens about 40% of the time.

Robustness & quality

Faults injected, causal features recomputed, scored on the same anchors. Quality score: completeness, recency, plausibility, rate of change.

Alerts & explainability

Alert-burden curves let a clinic trade sensitivity against false alarms explicitly. SHAP attribution over the fused feature set explains each forecast.

5 · CLINICAL OUTPUTS

Forecast + band

Trajectory to +120 min with its 80% interval and one plain sentence. The band widens with the horizon — that widening is the content.

Adverse-event risk

Calibrated chance of hyperglycaemia or hypoglycaemia inside the window, at the threshold the clinic chose. Thresholds follow the 2019 consensus.

What-if answers

Counterfactual meal size, carbohydrate, timing, an Indian preset swap, and a post-meal walk — through the monotone model and the twin.

Trust surface

Sensor data quality, personalisation depth, input tier available, feature attribution, and the limits printed beside each answer.

6 · INTERFACES

Web console `web/`

A single-page clinician console. Patient list with triage ordering, then per patient: forecast, what-if lab, glucose report (AGP, time in range, GMI per the 2019 consensus), health record, sleep and activity context, visit summary, and an evidence section carrying every cohort-level result.

Role-gated access — clinician and care coordinator see different fields — with a session access log. Dark and light themes, mg/dL ⇄ mmol/L, readable at tablet width.

Streamlit application `make app`

Ten pages: Overview, Live twin, Glucose report, What-if lab, Twin state, Personalization, Input tiers, Data quality, Evidence, and a PDF export for the patient file.

Both interfaces read the same bundle, so the two never disagree on a number. Every page carries the decision-support disclaimer: the clinician remains the decision maker and the system never recommends a dose or a drug.

METHOD — HOW EVERY NUMBER WAS OBTAINED

- **Pre-registered.** Each protocol, candidate set and decision rule written to `docs/DECISIONS.md` before the run that tested it. Over 70 numbered entries.
- **Paired patient bootstrap.** Intervals computed over patients, never over time windows — readings from one patient are not independent.
- **Untouched holdout.** Final claims confirmed on patients never used for any choice; the runner refuses a second run.
- **Null results published** at full size rather than dropped.

EVALUATION PROTOCOL

- **Patient-grouped CV** with repeats, an inner validation split for every choice, and a reserved holdout.
- **Baselines it must beat:** persistence, ridge regression, a GRU sequence model.
- **Horizons** +30, +60, +120 min, scored on mean absolute error.
- **Also scored:** interval coverage, event AUROC, alert burden, plausibility checks, 87 matched breakfasts.
- **Stress:** cross-cohort transfer, injected sensor faults, four hybrid variants, a synthetic Indian cohort.

MEASURED RESULTS

METRIC	VALUE
MAE at +30 min	**11.6**
MAE at +60 min	**16.8**
MAE at +120 min	**21.2**
AUROC, above 180	**0.82**
AUROC, below 70	**0.81**
Interval coverage	**80.8%**
Transfer gap	**3.4**

PRIVACY BY DESIGN

- Anonymised open or synthetic data only; no real patient record is in the repository.
- **Federated:** FedAvg matches pooled training (difference 0.09–0.12 mg/dL, intervals include zero) and beats a lone small clinic by ≈2 mg/dL.
- Data minimisation by design: only model updates are exchanged. No formal differential-privacy guarantee is claimed at this clinic count.
- Role-gated access with a session log; mapped to India's **DPDP Act** in `docs/limitations.md`.

ENGINEERING & SCOPE

- Python 3.11 · LightGBM · PyTorch · SHAP · Streamlit. Runs on a laptop CPU; no GPU required.
- One `make` target per stage; every figure and table regenerates from source. CI runs ruff, black, pytest and a smoke run on each push. Docker image, datasheet and model card included.
- One condition: **type 2 diabetes**. Decision support only — not a medical device.
- India, sleep and federated results are **simulations**: they test design, not clinical performance. Not prospectively validated.

Code MIT. Datasets keep their own licences. Simulated streams are labelled SIMULATED throughout.